3 A student sets up the apparatus illustrated in Fig. 3.1 in order to observe two-source interference fringes. The double slit with slit separation 0.800 mm, situated 2.50 m from the screen, is illuminated with coherent red light of wavelength 690 nm. Fringes are observed on the screen.

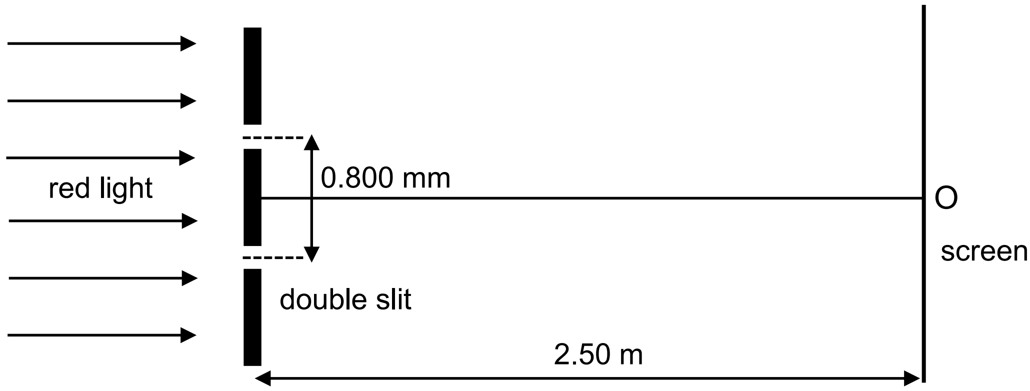

Fig. 3.1

(a) State two conditions necessary for two source interference fringes to be observed.

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Solution

Waves must be of the <u>same nature</u> and must <u>overlap</u>. B1

Sources must be <u>coherent</u>, i.e. the phase difference between them remains constant. B1

Amplitudes of the two waves must be approximately <u>equal</u>.

For transverse waves, they must be either <u>unpolarized</u> or, if polarized, <u>polarized in the same direction</u>.

(b) Explain why a maxima is always observed at Point O.

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Solution

The two sources are <u>in phase</u> and there is <u>no path difference</u> of the waves at Point O as it is equidistant from the two sources. B1

Therefore, <u>constructive interference</u> must occur as the two waves are in phase with each other at point O, giving rise to a maxima at point O. B1

(c) Calculate the distance from O to the second minima observed on the screen.

separation = …………………………………………… m [3]

Solution

$x = \dfrac{\lambda D}{a}$

$= \dfrac{(690 \times 10^{-9})(2.50)}{0.800 \times 10^{-3}}$ M1

$= 2.15625 \times 10^{-3}$ m

Distance from O to the second minima observed on the screen

$= 1.5x$

$= 1.5 \times (2.15625 \times 10^{-3})$ M1

$= 3.23 \times 10^{-3}$ m A1

(d) Describe the changes, if any, that occur in the separation of the fringes and the difference in the brightness between bright and dark fringes observed on the screen, when each of the following changes is made separately.

(i) increasing the intensity of the red light incident on the double slit,

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Solution

There is no change in the spacing. B1

The maxima is brighter and thus the difference in the brightness is higher. B1

(ii) increasing the distance between the double slit and the screen.

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Solution

Since D is larger, x will be larger. The separation of the fringes will increase. B1

The maxima is dimmer/ less intense/ less bright due to the longer distance travelled by the waves. Thus, the difference in the brightness is lower. B1

[Total: 11]